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(54) **SPINTRONIC DEVICE, MEMORY CELL, MEMORY ARRAY AND READ AND WRITE CIRCUIT**

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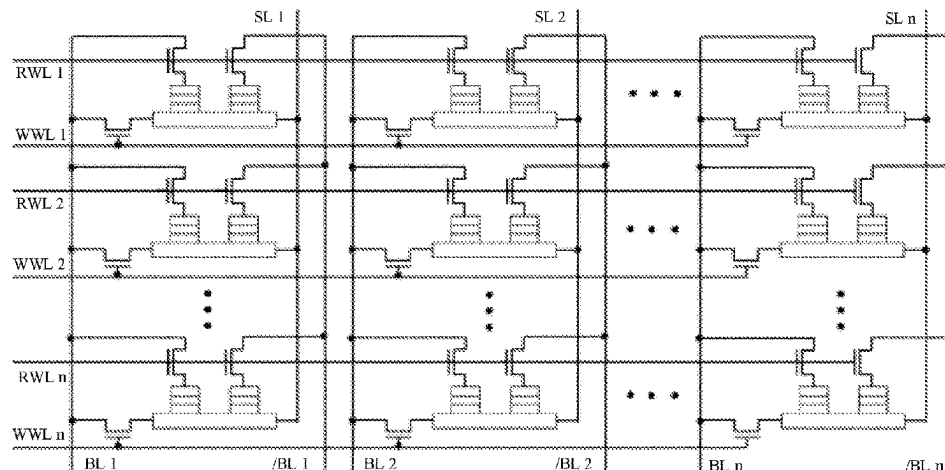
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(57) **ABSTRACT**

Provided is a spintronic device, a memory cell, a memory array, and a read and write circuit applied in a field of integration technology. The spintronic device includes: a bottom electrode; a spin orbit coupling layer, arranged on the bottom electrode; at least one pair of magnetic tunnel junctions, arranged on the spin orbit coupling layer, wherein each of the magnetic tunnel junctions includes a free layer, a tunneling layer, and a reference layer arranged sequentially from bottom to top, and wherein magnetization directions of reference layers of two magnetic tunnel junctions of each pair of the magnetic tunnel junctions are opposite; and a top

(Continued)



electrode, arranged on a reference layer of each of the magnetic tunnel junctions.

6 Claims, 7 Drawing Sheets

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- (52) **U.S. Cl.**
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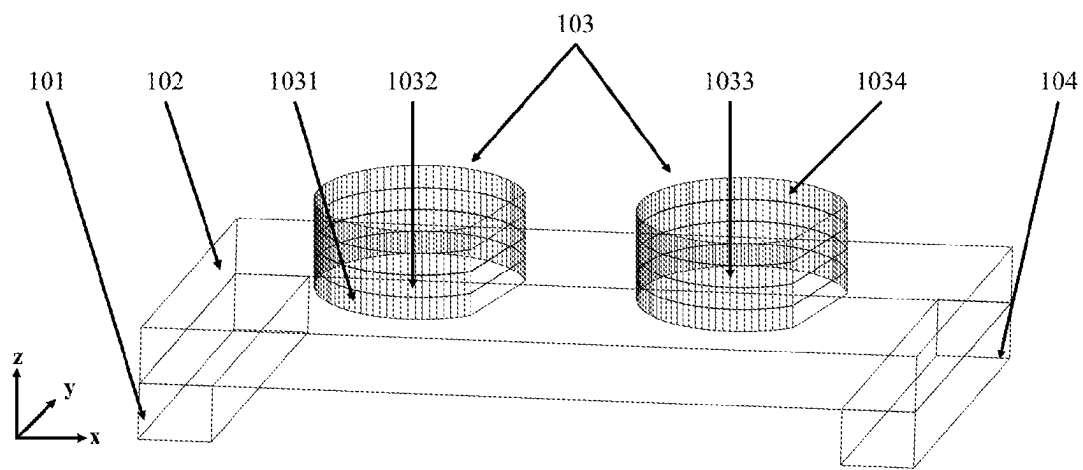


FIG. 1

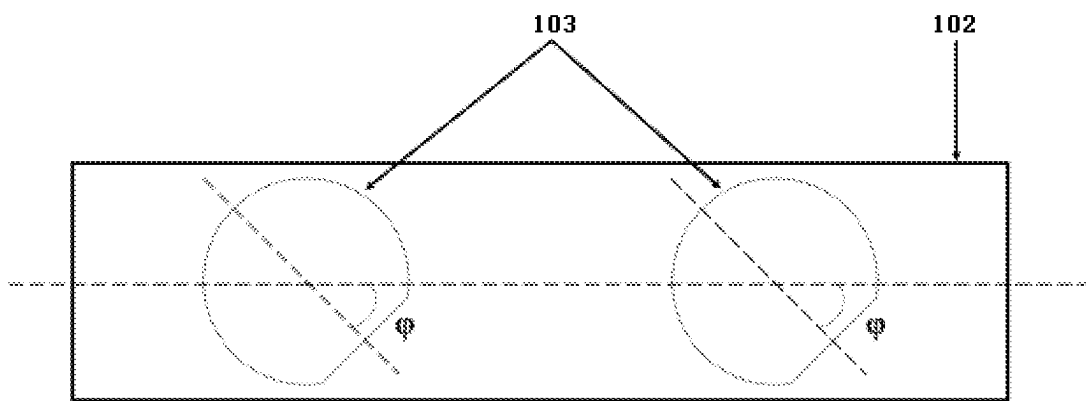


FIG. 2

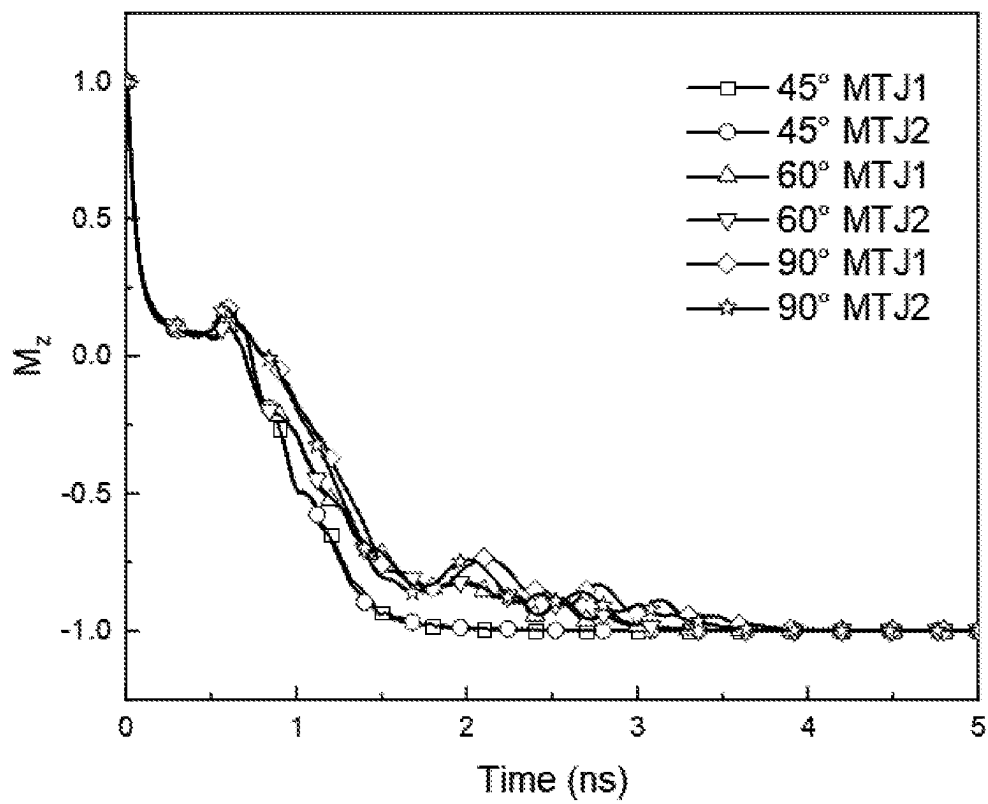


FIG. 3

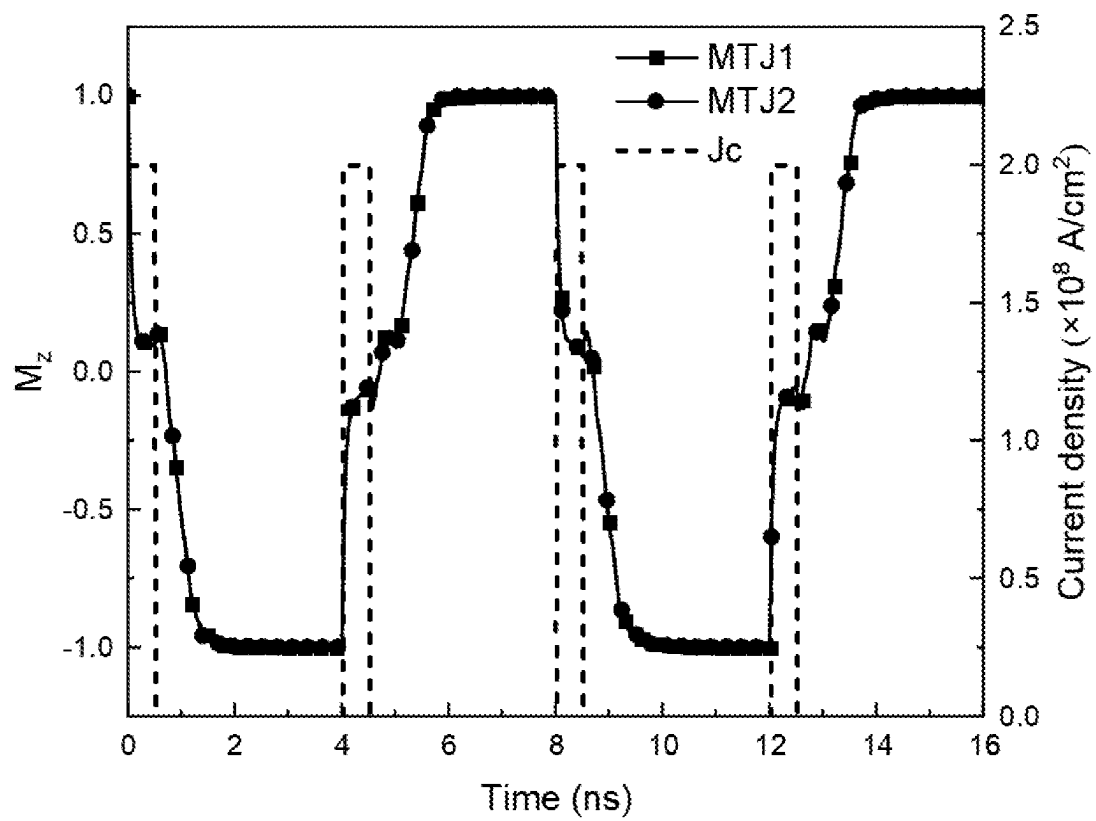


FIG. 4

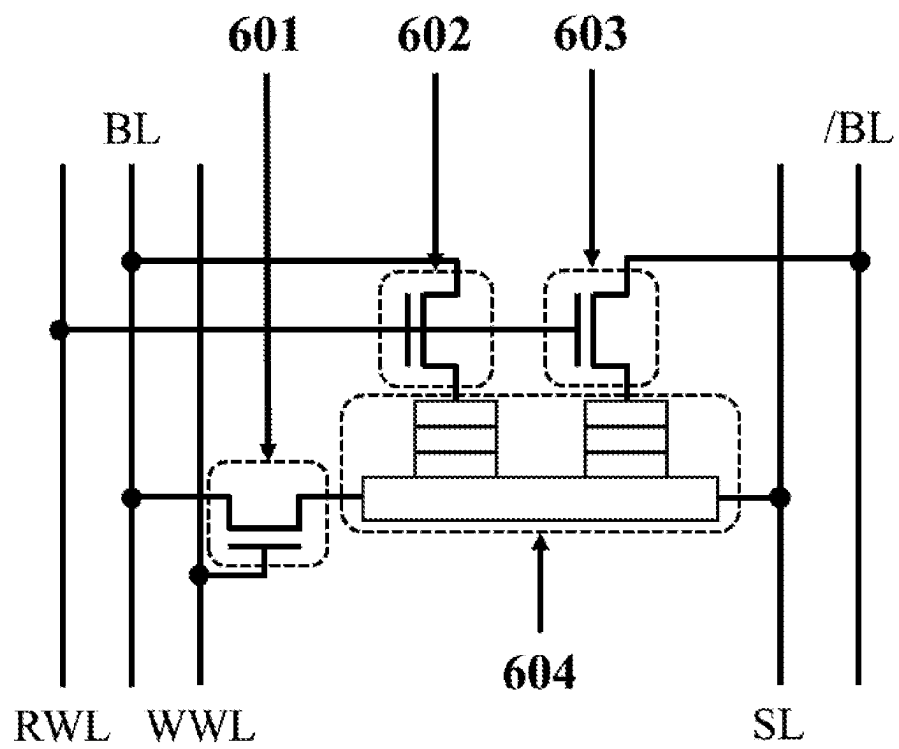


FIG. 5

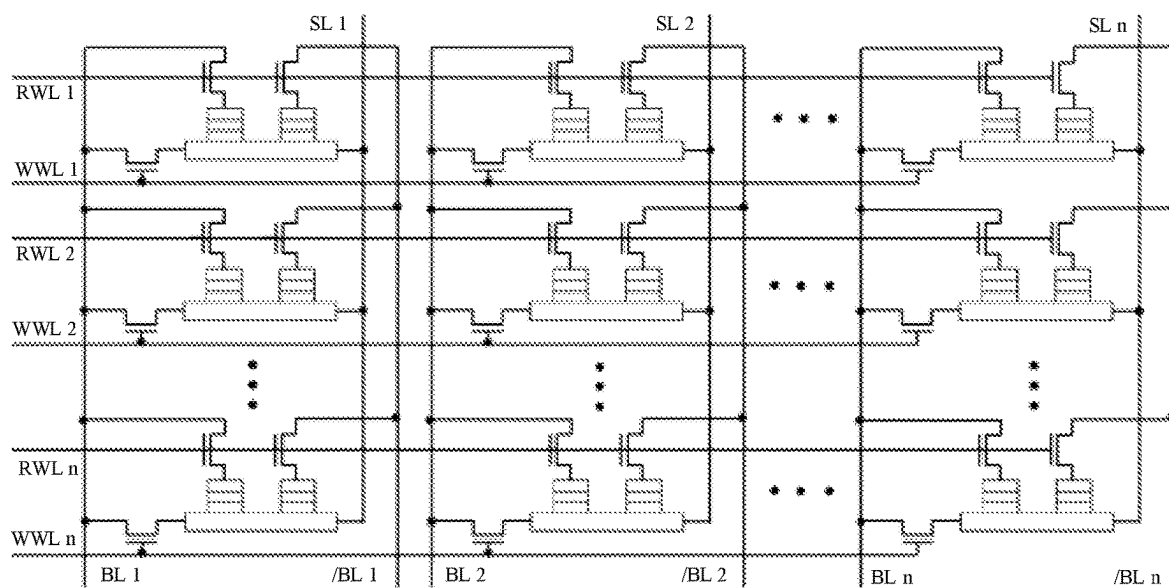


FIG. 6

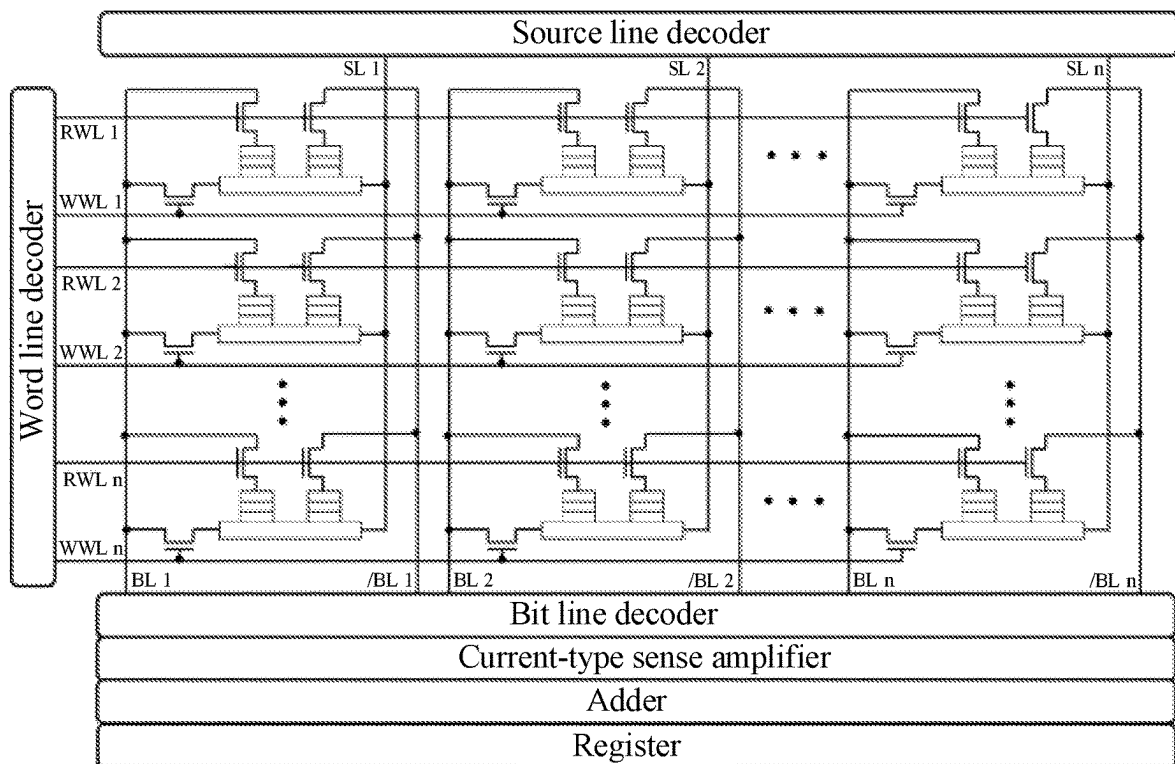


FIG. 7

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SPINTRONIC DEVICE, MEMORY CELL, MEMORY ARRAY AND READ AND WRITE CIRCUIT

CROSS-REFERENCE TO RELATED APPLICATION

This application is a Section 371 National Stage Application of International Application No. PCT/CN2021/123529, filed on Oct. 13, 2021, entitled "SPINTRONIC DEVICE, MEMORY CELL, MEMORY ARRAY AND READ AND WRITE CIRCUIT", the content of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

The present disclosure relates to a field of integrated circuits, and in particular, to a spintronic device, a memory cell, a memory array, and a read and write circuit.

BACKGROUND

A spin orbit torque magnetic random access memory (SOT-MRAM) switches a magnetization direction of a free layer through a spin current in a vertical direction induced by a current flowing through a heavy metal layer, and respectively achieves a low resistance state and a high resistance state according to the parallel or antiparallel arrangement of magnetization directions of the free layer and the reference layer. Meanwhile, the SOT-MRAM has the advantages of high speed and high endurance, which provides the possibility for integrated storage and computing applications. However, the switching of the traditional SOT-MRAM requires an assistance of an in-plane field, which is not conducive to integration. Moreover, a tunneling magnetoresistance (TMR) ratio of a magnetic tunnel junction (MTJ) is low, and generally only reaches 100%-150%, which makes a process fluctuation lead to a greater probability of misreading.

SUMMARY

The main objective of the present disclosure is to provide a spintronic device, a memory cell, a memory array, and a read and write circuit without an external field switching and with a high read and write margin.

In order to achieve the above objective, according to a first aspect of the embodiments of the present disclosure, there is provided a spintronic device, including:

a bottom electrode;
a spin orbit coupling layer arranged on the bottom electrode;

at least one pair of magnetic tunnel junctions arranged on the spin orbit coupling layer, wherein each of the magnetic tunnel junctions includes a free layer, a tunneling layer, and a reference layer arranged sequentially from bottom to top, and magnetization directions of reference layers of two magnetic tunnel junctions of each pair of the magnetic tunnel junctions are opposite; and

a top electrode arranged on a reference layer of each of the magnetic tunnel junctions.

In an embodiment, a structure of each of the magnetic tunnel junctions is a string-shaped structure.

In an embodiment, an axis of a straight side of each of the magnetic tunnel junctions forms a preset included angle with an axis of a length direction of the spin orbit coupling layer.

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In an embodiment, the bottom electrode includes:

a first electrode, connected to a first end of the spin orbit coupling layer; and

a second electrode, connected to a second end of the spin orbit coupling layer;

wherein the first end and the second end of the spin orbit coupling layer are arranged opposite to each other.

In an embodiment, a material of the spin orbit coupling layer is a heavy metal material.

According to a second aspect of the embodiments of the present application, there is provided a memory cell, including:

the spintronic device according to the first aspect of the embodiments of the present application;

a first transistor, including a first terminal, a second terminal and a third terminal, wherein the first terminal of the first transistor is connected to one end of the bottom electrode, the second terminal of the first transistor is connected to a write word line, the third terminal of the first transistor is connected to a bit line, and the other end of the bottom electrode is connected to a source line;

a second transistor, including a first terminal, a second terminal and a third terminal, wherein the first terminal of the second transistor is connected to one top electrode for a pair of magnetic tunnel junctions, the second terminal of the second transistor is connected to a read word line, and the third terminal of the second transistor is connected to the bit line; and

a third transistor, including a first terminal, a second terminal and a third terminal, wherein the first terminal of the third transistor is connected to another top electrode for the pair of magnetic tunnel junctions, the second terminal of the third transistor is connected to the read word line, and the third terminal of the third transistor is connected to an inverse bit line.

According a third aspect of the present disclosure, there is provided a memory array, including: m write word lines, m read word lines, n source lines, and m rows and n columns of memory cells, wherein the memory cell is the memory cell according to the second aspect of the embodiments of the present disclosure, each of m and n is a positive integer;

a bit line for each of the memory cells in a same column is connected to a same bit line, and an inverse bit line for each of the memory cells in a same column is connected to a same inverse bit line; and

a write word line for each of the memory cells in a same row is connected to a same write word line, and a read word line for each of the memory cells in a same column is connected to a same read word line.

According to a fourth aspect of the embodiments of the present disclosure, there is provided a read and write circuit, including:

the memory array according to the third aspect of the embodiments of the present disclosure;

a bit line decoder, configured to provide a bit line operating voltage to n bit lines and n inverse bit lines;

a word line decoder, configured to provide a word line operating voltage to m write word lines and m read word lines;

a source line decoder, configured to provide a bit line operating voltage and an induced current to n source lines; and

a reading operation module, configured to read data stored in the memory array and perform a logical operation on the data stored in the memory array.

In an embodiment, the read operation module includes: a current-type sense amplifier, including an input end, a reference end and an output end, wherein the input end of the current-type sense amplifier is connected to the bit line through the bit line decoder, and the reference end of the current-type sense amplifier is connected to the inverse bit line through the bit line decoder; an adder, including an input end and an output end, wherein the input end of the adder is connected to the output end of the current-type sense amplifier; and a register, connected to the output end of the adder.

In an embodiment, the read and write circuit is configured for a binary neural network.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic structural diagram of a spintronic device according to an embodiment of the present disclosure.

FIG. 2 is a top view of a spintronic device according to an embodiment of the present disclosure.

FIG. 3 is a magnetization switching curve of a spintronic device according to an embodiment of the present disclosure.

FIG. 4 is a synchronous switching curve of a unipolar pulse of a spintronic device according to an embodiment of the present disclosure.

FIG. 5 is a schematic diagram of a memory cell according to an embodiment of the present disclosure.

FIG. 6 is a schematic diagram of a memory array according to an embodiment of the present disclosure.

FIG. 7 is a schematic diagram of a read and write circuit according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

In order to make the objectives, features and advantages of the present disclosure more obvious and understandable, the technical solutions in the embodiments of the present disclosure will be clearly and completely described below in combination with the drawings in the embodiments of the present disclosure. Obviously, the described embodiments are only a portion of the embodiments of the present disclosure, but not all thereof. Based on the embodiments in the present disclosure, all other embodiments obtained by those skilled in the art without creative efforts belong to the protection scope of the present disclosure.

Referring to FIG. 1, FIG. 1 is a schematic structural diagram of a spintronic device according to an embodiment of the present disclosure. The spintronic device includes: a bottom electrode **101**, **104**; a spin orbit coupling layer **102**, arranged on the bottom electrode **101**, **104**; at least one pair of magnetic tunnel junctions **103**, arranged on the spin orbit coupling layer **102**, wherein each of the magnetic tunnel junctions **103** comprises a free layer **1031**, a tunneling layer **1032**, and a reference layer **1033** arranged sequentially from bottom to top, and magnetization directions of reference layers **1033** of each pair of the magnetic tunnel junctions **103** are opposite; and a top electrode **1034**, arranged on a reference layer **1033** of each of the magnetic tunnel junctions **103**.

In an embodiment of the present disclosure, materials of the reference layer **1033** and the free layer **1031** are one or more of ferromagnetic materials having perpendicular magnetic anisotropy such as CoFeB and Co/Pt. The spin orbit coupling layer **102** is composed of one or more of heavy metals such as Pt, Ta, and W. The bottom electrode **101**, **104**

and the top electrode **1034** are composed of metals such as Ti/Au, Ti/Pt, Cr/Au and Ta/CuN. Furthermore, the tunneling layer **1032** is composed of MgO, Al₂O₃ and the like.

Referring to FIG. 2, FIG. 2 is a top view of a spintronic device according to an embodiment of the present disclosure.

In an embodiment of the present disclosure, a structure of each of the magnetic tunnel junctions **103** is a string-shaped structure, and an axis of a straight side of each of the magnetic tunnel junctions **103** forms a preset included angle φ with an axis of a length direction of the spin orbit coupling layer **102**. Such asymmetric shape of the magnetic tunnel junction **103** will induce a certain shape anisotropy, combined with the DM antisymmetric exchange interaction, a field-free deterministic switching of the magnetic tunnel junction **103** may be implemented and a unipolar switching may also be implemented. When a current flows through the spin orbit coupling layer **102**, due to the spin Hall effect of a heavy metal or the Rashba effect of an interface, a spin current in a vertical direction will be generated, so that the free layer **1031** of the magnetic tunnel junction **103** is switched. As the magnetization directions of the reference layers **1033** of two magnetic tunnel junctions **103** are opposite, the two magnetic tunnel junctions **103** always store opposite resistance states.

In the present disclosure, preset included angles φ formed between axes of straight sides of two magnetic tunnel junctions **103** in a pair of magnetic tunnel junctions and the axis of the length direction of the spin orbit coupling layer **102** may be the same or complementary, which is not limited in the present disclosure.

Referring to FIG. 3, FIG. 3 shows a magnetization switching curve of a spintronic device according to an embodiment of the present disclosure. Preset included angle φ of a pair of magnetic tunnel junctions **103** is sequentially set to be 45°, 60° and 90°, and a current pulse with a pulse width of 0.5 ns and a density of 2×10^8 A/cm² is applied along an x-axis direction, thereby achieving a synchronous field-free deterministic switching of the pair of magnetic tunnel junctions **103**.

Referring to FIG. 4, FIG. 4 shows a synchronous switching curve of a unipolar pulse of a spintronic device according to an embodiment of the present disclosure. A preset included angle φ of a pair of magnetic tunnel junctions **103** is set to be 45°, and four unipolar current pulses with a pulse width of 0.5 ns and a density of 2×10^8 A/cm² are continuously applied along the x-axis direction, thereby achieving a synchronous repeated switching of two magnetic tunnel junctions **103**. A unipolar current pulse is easy to operate and simplifies a peripheral driving circuit.

Referring to FIG. 5, FIG. 5 is a schematic diagram of a memory cell according to an embodiment of the present disclosure. The memory cell includes: the spintronic device **604** as shown in FIG. 1; a first transistor **601**, including a first terminal, a second terminal and a third terminal, wherein the first terminal of the first transistor is connected to the bottom electrode **101**, the second terminal of the first transistor is connected to a write word line WWL, the third terminal of the first transistor **601** is connected to a bit line BL, and the bottom electrode **104** is connected to a source line SL; a second transistor **602**, including a first terminal, a second terminal and a third terminal, wherein the first terminal of the second transistor **602** is connected to one top electrode **1034** for a pair of magnetic tunnel junctions **103**, the second terminal of the second transistor **602** is connected to a read word line RWL, and the third terminal of the second transistor **602** is connected to the bit line BL; and a third

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transistor **603**, including a first terminal, a second terminal and a third terminal, wherein the first terminal of the third transistor **603** is connected to another top electrode **1034** for the pair of magnetic tunnel junctions, the second terminal of the third transistor **603** is connected to the read word line RWL, and the third terminal of the third transistor **603** is connected to an inverse bit line/BL.

In this embodiment, **601** may be regarded as a write control transistor, **602** and **603** may be regarded as read control transistors, and **604** is a spintronic device. When a reading operation is operated, the write word line WWL is pulled low to turn off the first transistor **601**, the read word line RWL is pulled high to turn on the second transistor **602** and the third transistor **603**, while the source line SL is grounded, and consequently the current flows through a pair of magnetic tunnel junctions **103** via the bit line BL and the inverse bit line/BL. The current on the bit line BL and the inverse bit line/BL flows into a current-type sense amplifier CSA to read a storage state of the spintronic device **604**. When a writing operation is performed, the write word line WWL is pulled high to turn on the first transistor **601**, the read word line RWL is pulled low to turn off the second transistor **602** and the third transistor **603** are turned off, while the source line SL is grounded, the current between the bit line BL and the source line SL flows through a current path of the spin orbit coupling layer **102**, and the resistance state of the magnetic tunnel junction **103** is switched, thereby achieving the writing of the resistance state.

Referring to FIG. 6, FIG. 6 is a schematic diagram of a memory array according to an embodiment of the present disclosure. The memory array includes: m write word lines, m read word lines, n source lines, and m rows and n columns of memory cells, wherein the memory cell is the memory cell as shown in FIG. 5, each of m and n is a positive integer; a bit line for each of the memory cells in a same column is connected to a same bit line, and an inverse bit line for each of the memory cells in a same column is connected to a same inverse bit line; and a write word line for each of the memory cells in a same row is connected to a same write word line, and a read word line for each of the memory cells in a same column is connected to a same read word line.

Referring to FIG. 7, FIG. 7 is a schematic diagram of a read and write circuit according to an embodiment of the present disclosure. The read and write circuit includes:

- the memory array as shown in FIG. 6;
- a bit line decoder, used to provide a bit line operating voltage to n bit lines and n inverse bit lines;
- a word line decoder, used to provide a word line operating voltage to m write word lines and m read word lines;
- a source line decoder, used to provide a bit line operating voltage and an induced current to n source lines; and
- a reading operation module, used to read data stored in the memory array and perform a logical operation on the data stored in the memory array.

In the present embodiment, at the algorithm level, the high resistance state and the low resistance state are respectively mapped to be (+1, 0). When a preceding stage input is a high level, according to the difference between the currents on the bit line BL and the inverse bit line/BL, a multiplication operation result may be output and input into an adder and a register; when the preceding stage input is a low level, no output will be generated, i.e., the output is a low level. After a series of multiplication and addition operations, a matrix-vector multiplication operation may be achieved for a binary neural network.

In an embodiment, the reading operation module includes: a current-type sense amplifier, including an input

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end, a reference end and an output end, wherein the input end of the current-type sense amplifier is connected to the bit line through the bit line decoder, and the reference end of the current-type sense amplifier is connected to the inverse bit line through the bit line decoder; an adder, including an input end and an output end, wherein the input end of the adder is connected to the output end of the current-type sense amplifier; and a register, connected to the output end of the adder.

Furthermore, it is also possible that the reference end of the current-type sense amplifier is connected to the bit line through the bit line decoder, and the input end of the current-type sense amplifier is connected with the inverse bit line through the bit line decoder, which is not limited by the present disclosure.

It may be known from the above embodiments of the present disclosure that the spintronic device, the memory cell, the memory array, and the read and write circuit provided by the present disclosure may, on one hand, achieve a deterministic magnetization switching driven by a unipolar current pulse without an external magnetic field. On another hand, the magnetization directions of the reference layers of two magnetic tunnel junctions of each pair of magnetic tunnel junctions are opposite, thereby ensuring that the two magnetic tunnel junctions are always in opposite resistance states, thereby achieving a self-reference and improving a reading margin. On yet another hand, the constructed memory array may achieve a matrix-vector multiplication operation in combination with an external circuit.

It should be noted that various functional modules in various embodiments of the present disclosure may be integrated into one processing module, or various modules may exist in a separately physical manner, or two or more modules may be integrated into one module. The above integrated modules may be implemented in the form of hardware or in the form of software functional modules.

If the integrated modules are implemented in the form of software functional modules and sold or used as independent products, they may be stored in a computer readable storage medium. Based on such an understanding, the essence of the technical solution of the present invention or the portion thereof that contributes to the prior art or all or a portion of the technical solution may be embodied in the form of software products.

It should be noted that, for the sake of simple description, various method embodiments described above are expressed as a series of action combinations, but those skilled in the art should know that the present disclosure is not limited by the described action sequences, because according to the present disclosure, certain steps may be performed in other sequences or simultaneously. Then, those skilled in the art should also know that the embodiments described in the specification belong to preferred embodiments, and the actions and modules involved are not necessarily required in the present disclosure.

In the above embodiments, the descriptions of various embodiments have their own emphases, and for portions not described in detail in a certain embodiment, reference may be made to relevant descriptions of other embodiments.

The above is a description of a spintronic device, a memory cell, a memory array, and a read and write circuit according to the present disclosure. Based on the concept of the embodiments of the present disclosure, those skilled in the art may make modifications to both the specific implementations and the application scopes. In summary, the contents of the present specification should not be construed as limiting the present disclosure.

What is claimed is:

1. A read and write circuit, comprising:

a memory array that comprises:

m write word lines, m read word lines, n source lines,
and m rows and n columns of memory cells, wherein
each of m and n is a positive integer, and each
memory cell comprises:

a spintronic device that comprises:

a bottom electrode;

a spin orbit coupling layer arranged on the bottom
electrode;

at least one pair of magnetic tunnel junctions
arranged on the spin orbit coupling layer,
wherein each of the magnetic tunnel junctions
comprises a free layer, a tunneling layer, and a
reference layer arranged sequentially from bot-
tom to top, and magnetization directions of the
reference layers of two the magnetic tunnel
junctions making up each pair of the magnetic
tunnel junctions are opposite; and

a top electrode, arranged on the reference layer of
each of the magnetic tunnel junctions;

a first transistor comprising a first terminal, a second
terminal and a third terminal, wherein the first
terminal of the first transistor is connected to the
bottom electrode, the second terminal of the first
transistor is connected to a write word line, the
third terminal of the first transistor is connected to
a bit line, and the bottom electrode is connected to
a source line;

a second transistor comprising a first terminal, a
second terminal and a third terminal, wherein the
first terminal of the second transistor is connected
to a first top electrode for the pair of magnetic
tunnel junctions, the second terminal of the second
transistor is connected to a read word line, and the
third terminal of the second transistor is connected
to the bit line; and

a third transistor comprising a first terminal, a second
terminal and a third terminal, wherein the first
terminal of the third transistor is connected to a
second top electrode for the pair of magnetic
tunnel junctions, the second terminal of the third
transistor is connected to the read word line, and
the third terminal of the third transistor is con-
nected to an inverse bit line;

wherein a bit line for each of the memory cells in a
same column is connected to a same bit line, and an
inverse bit line for each of the memory cells in a
same column is connected to a same inverse bit line;
and

wherein a write word line for each of the memory cells
in a same row is connected to a same write word line,
and a read word line for each of the memory cells in
a same column is connected to a same read word
line;

a bit line decoder, configured to provide a bit line oper-
ating voltage to n bit lines and n inverse bit lines;

a word line decoder, configured to provide a word line
operating voltage to m write word lines and m read
word lines;

a source line decoder, configured to provide a bit line
operating voltage and an induced current to n source
lines; and

a reading operation module, configured to read data stored
in the memory array and perform a logical operation on
the data stored in the memory array;

wherein the read and write circuit is configured for a
binary neural network.

2. The read and write circuit according to claim 1, wherein
in the spintronic device of each memory cell, a structure of
each of the magnetic tunnel junctions is a string-shaped
structure.

3. The read and write circuit according to claim 2, wherein
in the spintronic device of each memory cell, an axis of a
straight side of each of the magnetic tunnel junctions forms
a preset included angle with an axis of a length direction of
the spin orbit coupling layer.

4. The read and write circuit according to claim 1, wherein
in the spintronic device of each memory cell, the bottom
electrode comprises:

a first electrode connected to a first end of the spin orbit
coupling layer; and

a second electrode connected to a second end of the spin
orbit coupling layer;

wherein the first end and the second end of the spin orbit
coupling layer are arranged opposite to each other.

5. The read and write circuit according to claim 1, wherein
in the spintronic device of each memory cell, a material of
the spin orbit coupling layer is a heavy metal material.

6. The read and write circuit according to claim 1, wherein
the reading operation module comprises:

a current-type sense amplifier comprising an input end, a
reference end and an output end, wherein the input end
of the current-type sense amplifier is connected to the
bit line through the bit line decoder, and the reference
end of the current-type sense amplifier is connected to
the inverse bit line through the bit line decoder;

an adder comprising an input end and an output end,
wherein the input end of the adder is connected to the
output end of the current-type sense amplifier; and
a register connected to the output end of the adder.

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